# MS1Vtx_l3hcal_raw

overview · all regions and selections · rank 95 / 169



rank 95 / 169  by endcap / ht40 / D_equal

| region | selection | D_equal | D_pooled | 125/15 | 125/55 | 200/50 | 400/100 | 600/150 | 1000/275 | n_bkg |
|---|---|---|---|---|---|---|---|---|---|---|
| endcap | all | 0.193 | 0.192 | 0.196 | 0.190 | 0.196 | 0.197 | 0.184 | 0.196 | 34,970 |
| endcap | ht40 | 0.238 | 0.238 | 0.241 | 0.235 | 0.241 | 0.242 | 0.229 | 0.241 | 7,390 |
| endcap | bkg split | 0.058 | below 27,580 / above 7,390 | | | | | | | |
| barrel | all | 0.240 | 0.236 | 0.265 | 0.235 | 0.236 | 0.245 | 0.239 | 0.224 | 1,089 |
| barrel | ht40 | 0.373 | 0.369 | 0.397 | 0.367 | 0.368 | 0.377 | 0.372 | 0.357 | 220 |
| barrel | bkg split | 0.183 | below 869 / above 220 | | | | | | | |
| all | all | 0.178 | 0.176 | 0.190 | 0.174 | 0.181 | 0.183 | 0.170 | 0.172 | 36,059 |
| all | ht40 | 0.226 | 0.224 | 0.238 | 0.222 | 0.228 | 0.231 | 0.217 | 0.220 | 7,610 |
| all | bkg split | 0.061 | below 28,449 / above 7,610 | | | | | | | |

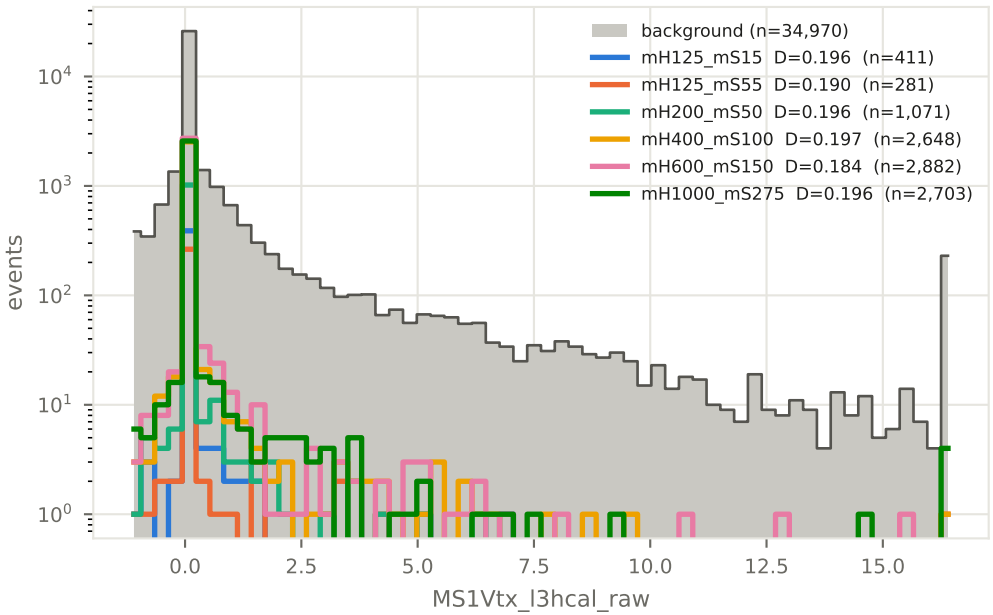
selections: all = every event;  ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

# MS1Vtx_l3hcal_raw

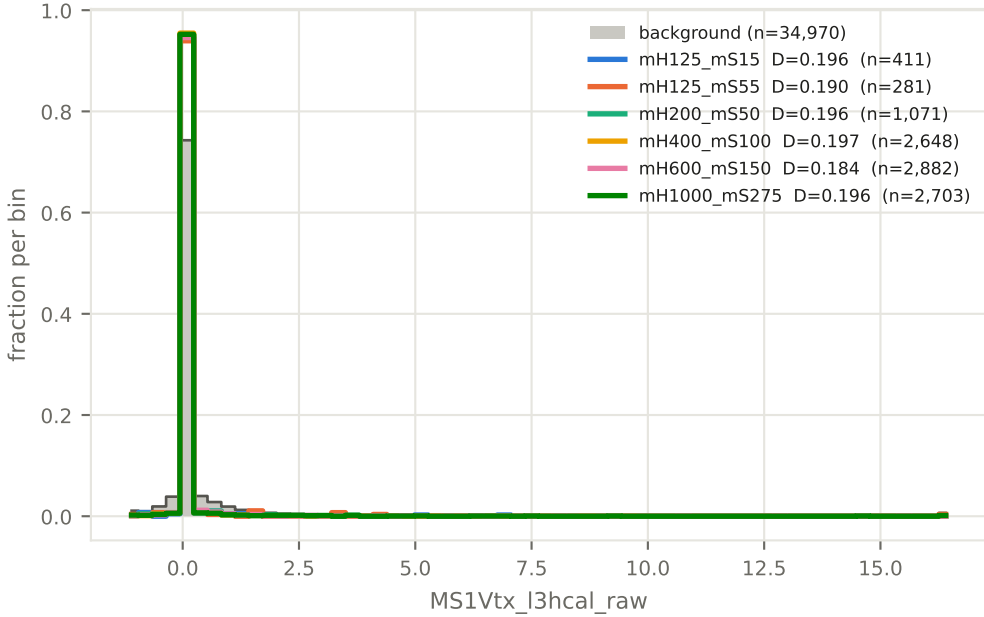
region: ENDCAP · selection: all (every event) · rank 95 / 169

D_equal 0.193 D_pooled 0.192 min 0.184 (mH600_mS150) max 0.197 (mH400_mS100) bkg n = 34,970 sig n = 9,996

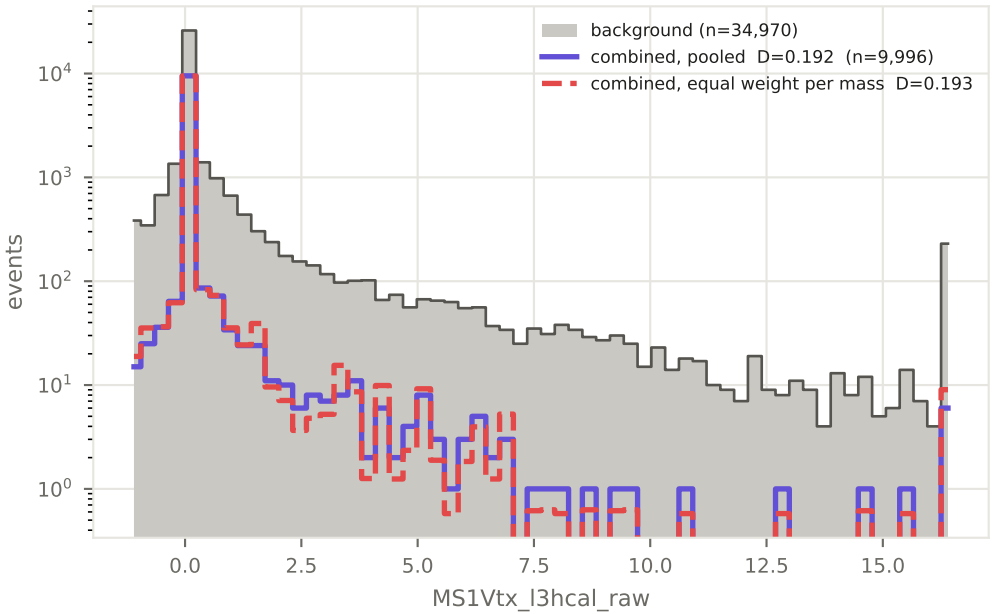
each mass point vs background · raw



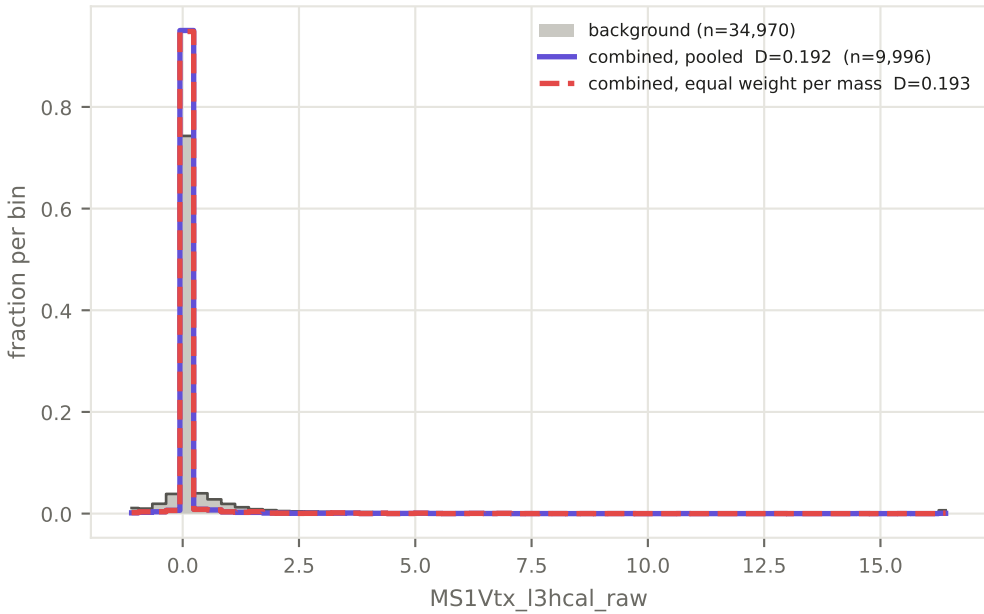

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized



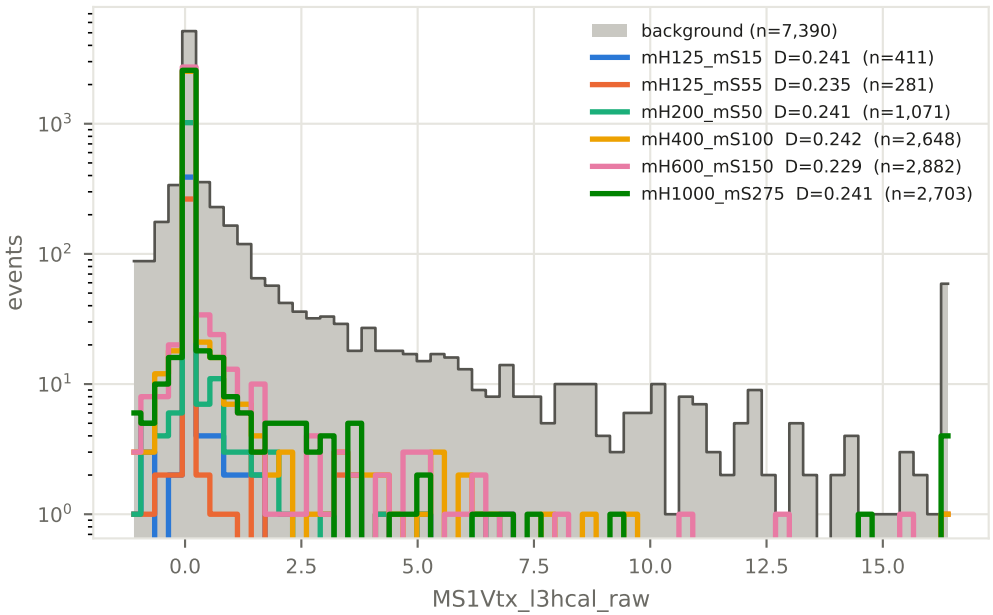

edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_l3hcal_raw

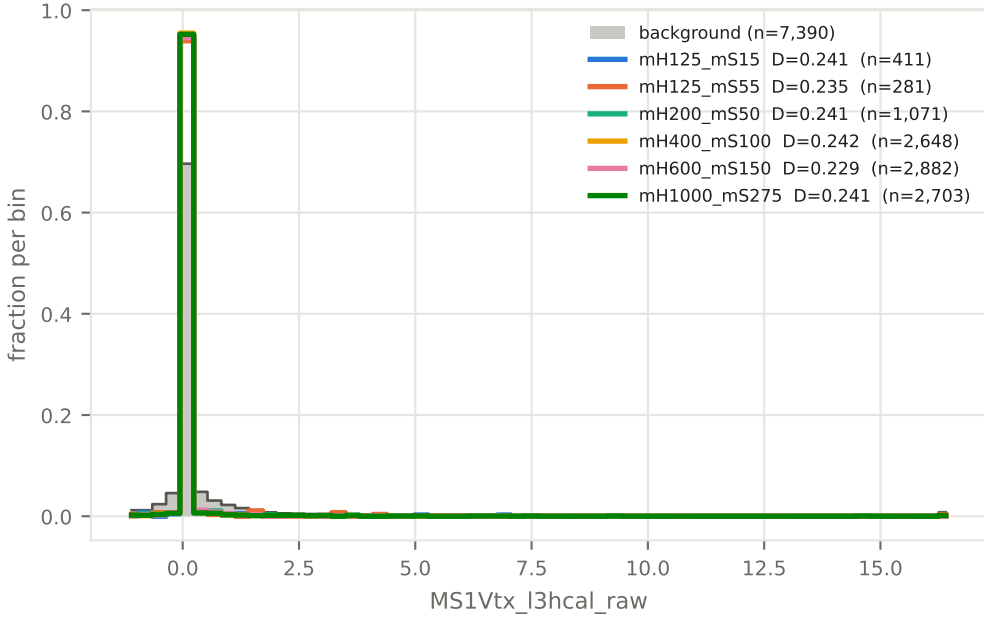
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 95 / 169

D_equal 0.238 D_pooled 0.238 min 0.229 (mH600_mS150) max 0.242 (mH400_mS100) bkg n = 7,390 sig n = 9,996

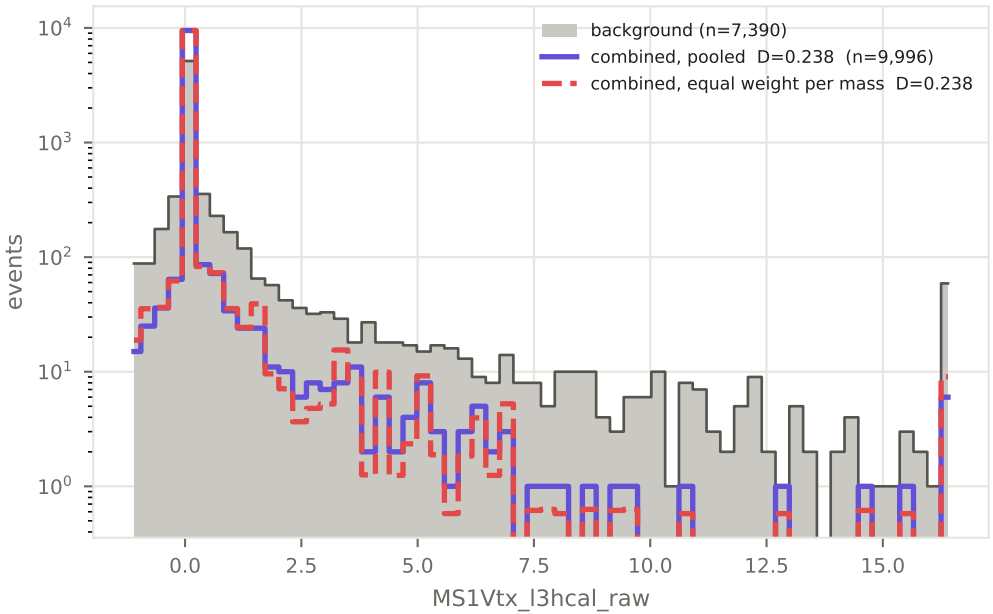
each mass point vs background · raw



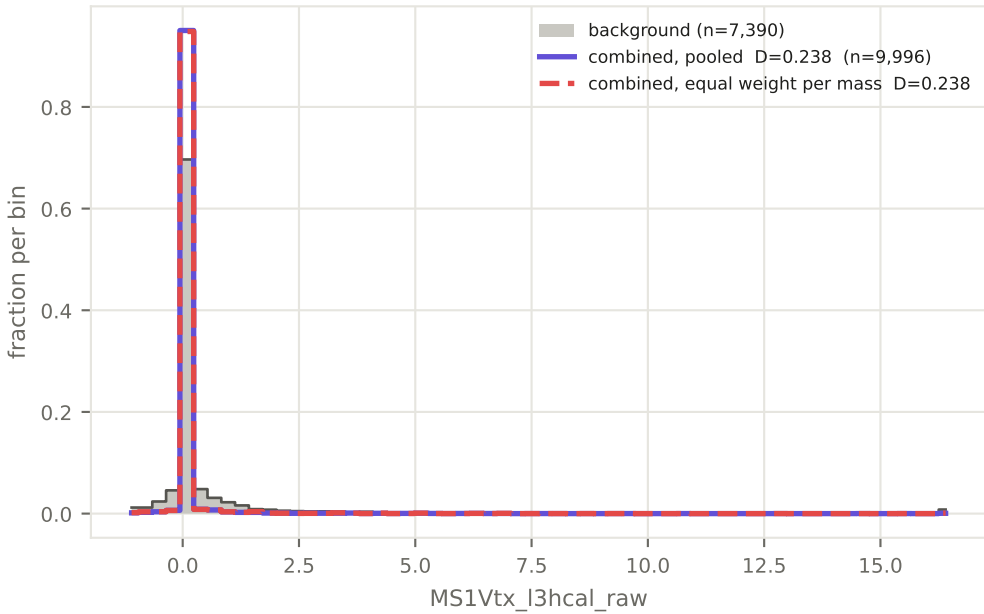

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized

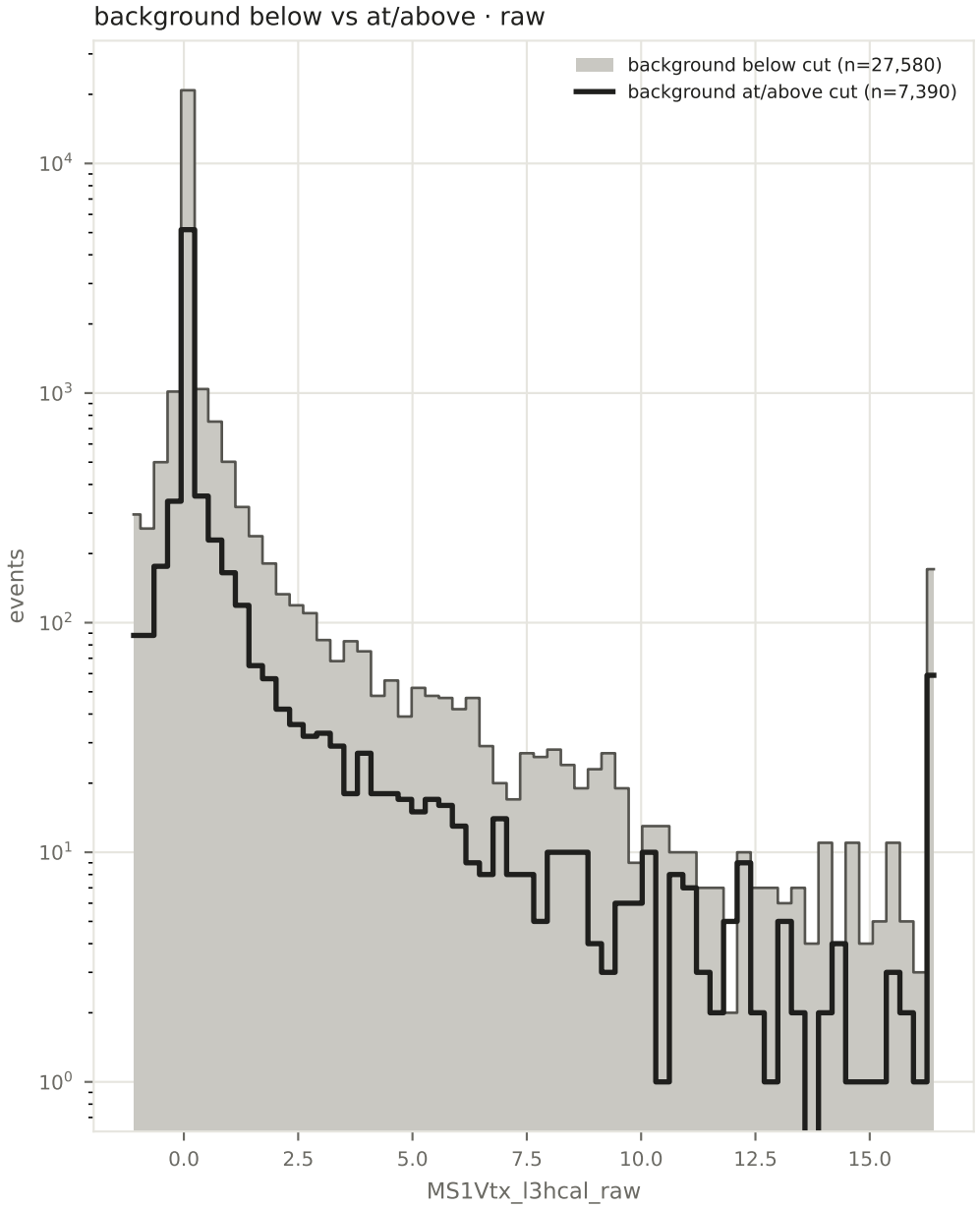


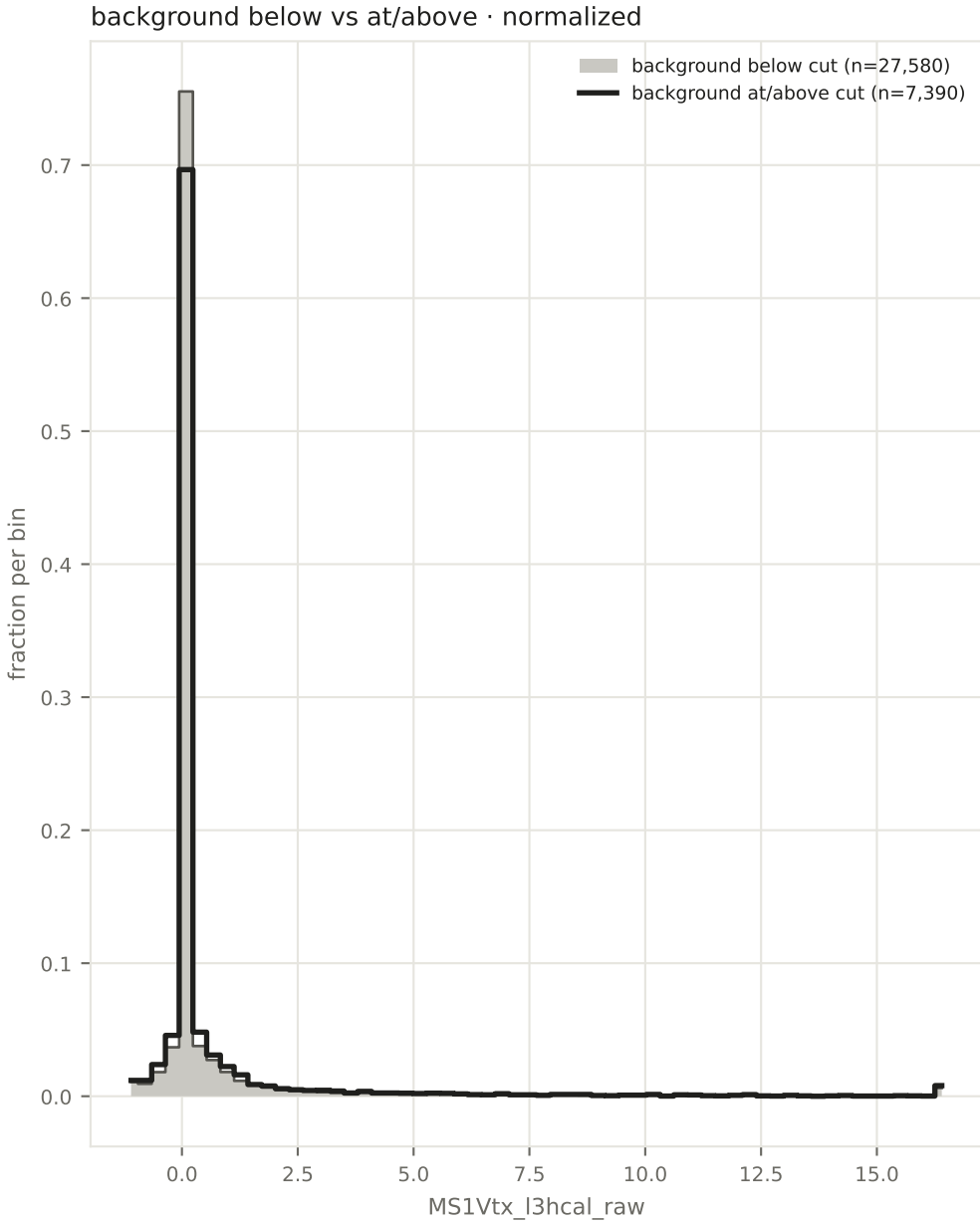

edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_l3hcal_raw

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 95 / 169

D(below vs above) 0.058  n below = 27,580  n above = 7,390  large D = background shape changes across the cut



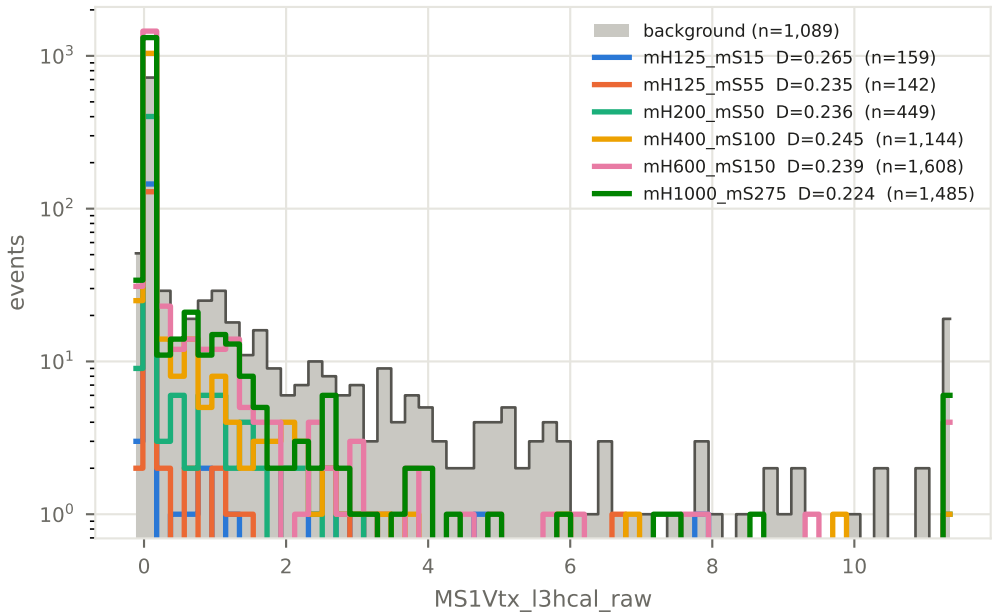



# MS1Vtx_l3hcal_raw

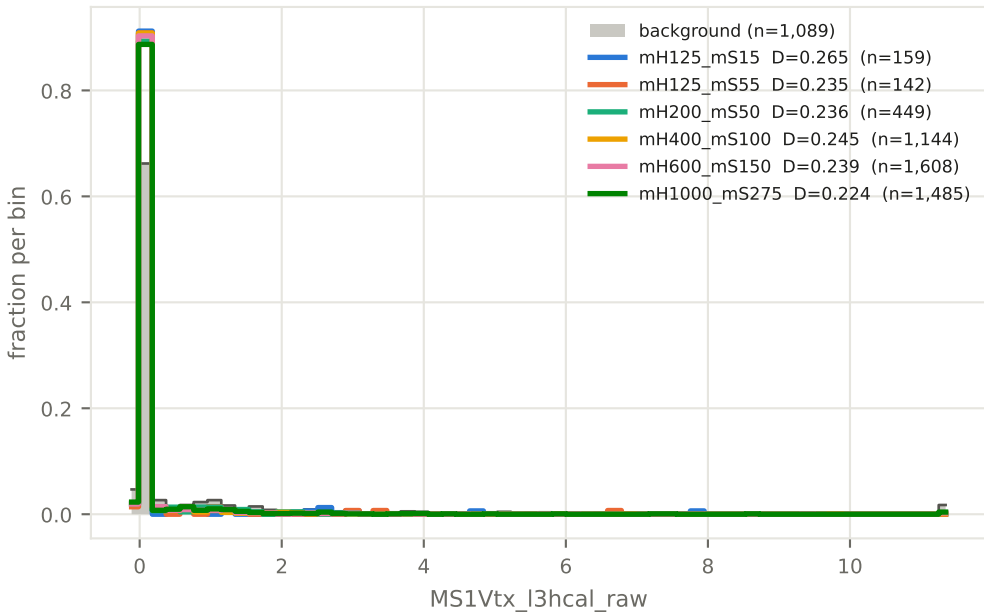
region: BARREL · selection: all (every event) · rank 95 / 169

D_equal 0.240 D_pooled 0.236 min 0.224 (mH1000_mS275) max 0.265 (mH125_mS15) bkg n = 1,089 sig n = 4,987

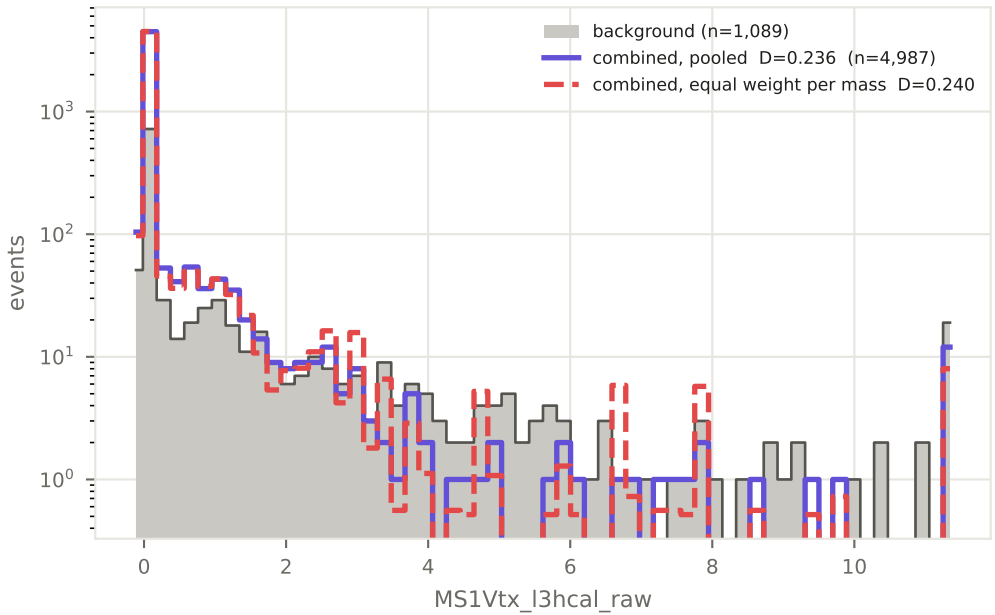
each mass point vs background · raw



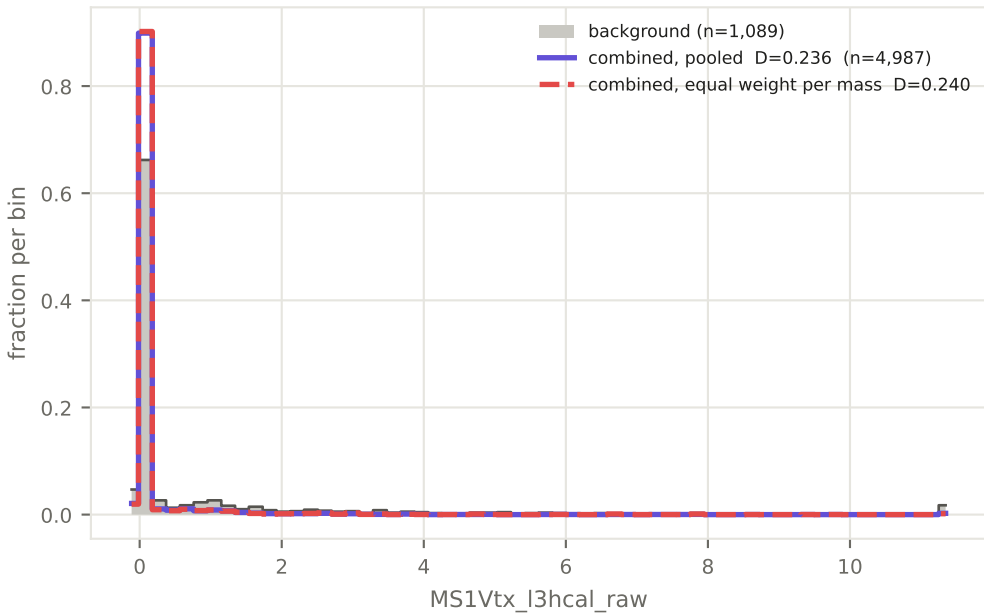

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized



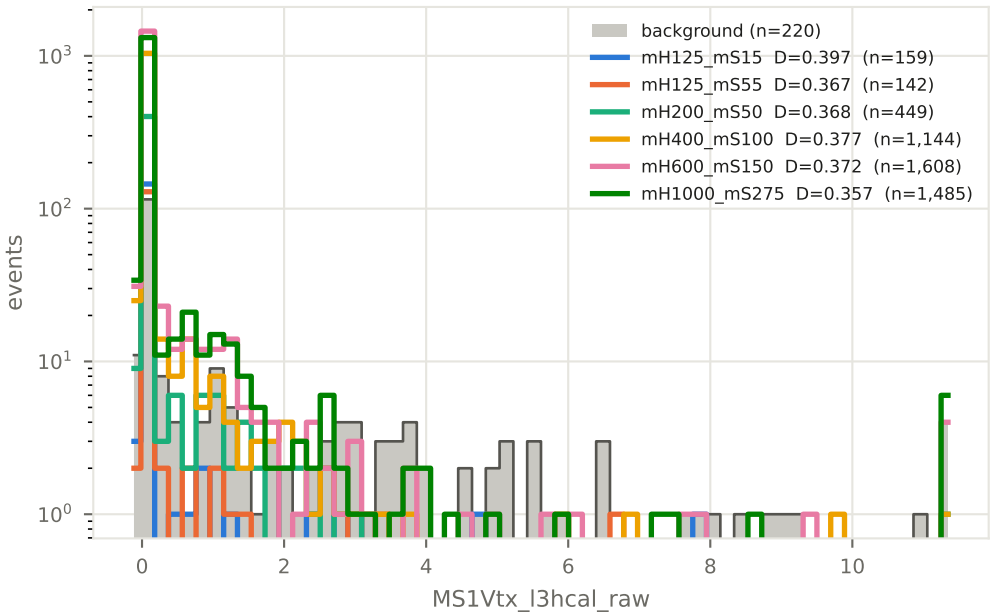

edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_l3hcal_raw

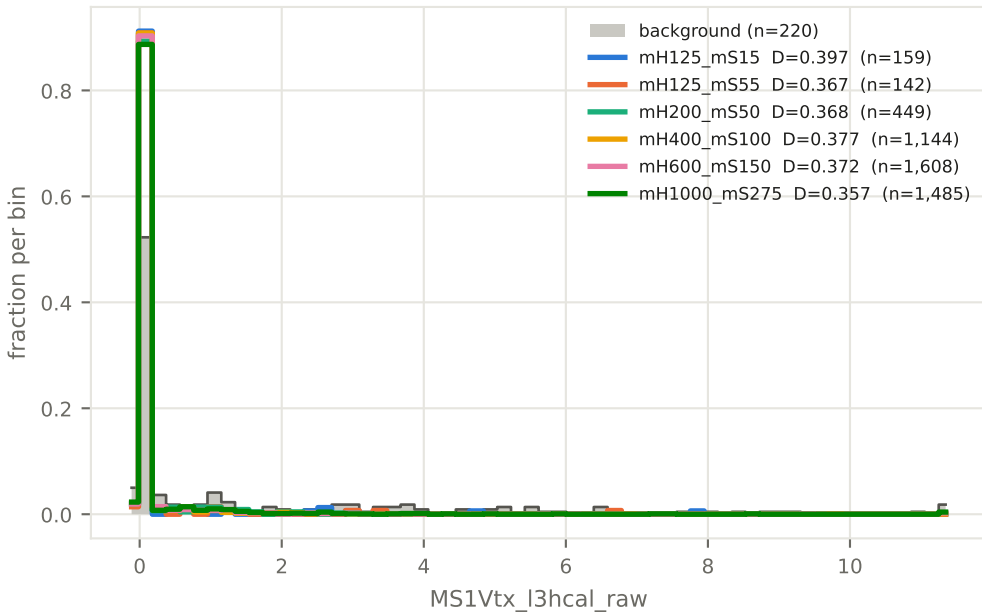
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 95 / 169

D_equal 0.373 D_pooled 0.369 min 0.357 (mH1000_mS275) max 0.397 (mH125_mS15) bkg n = 220 sig n = 4,987

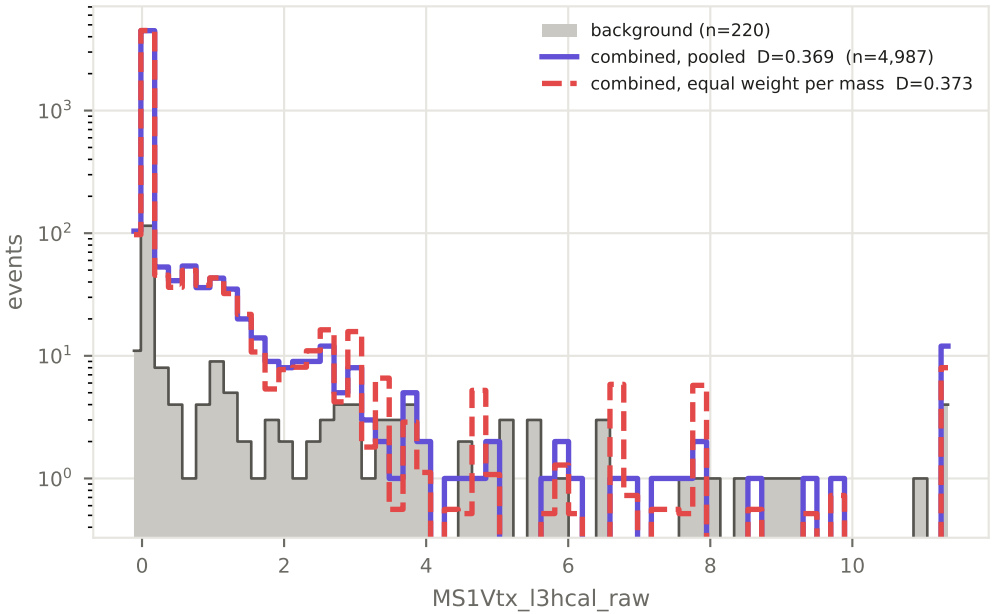
each mass point vs background · raw



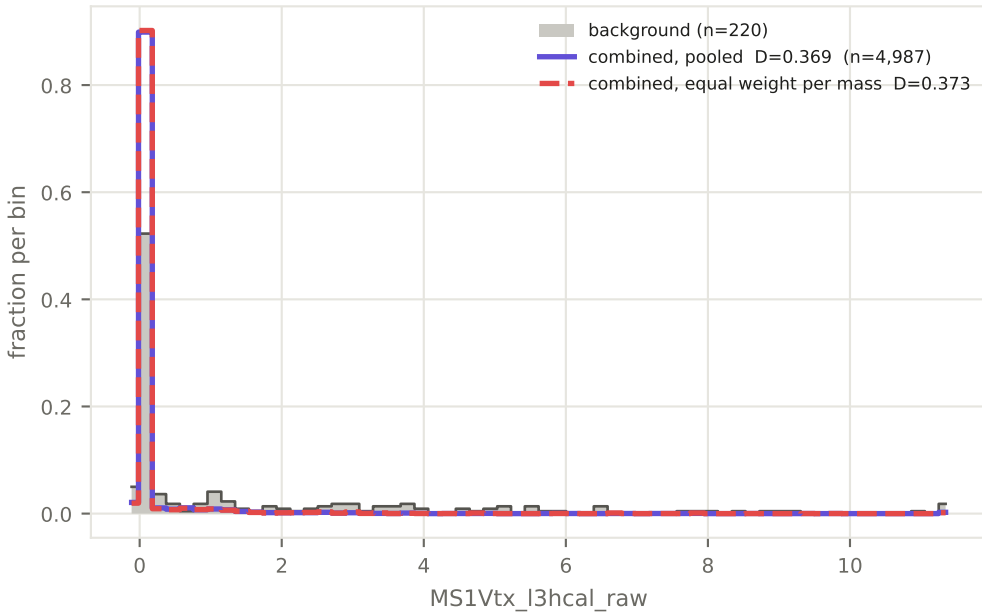

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized

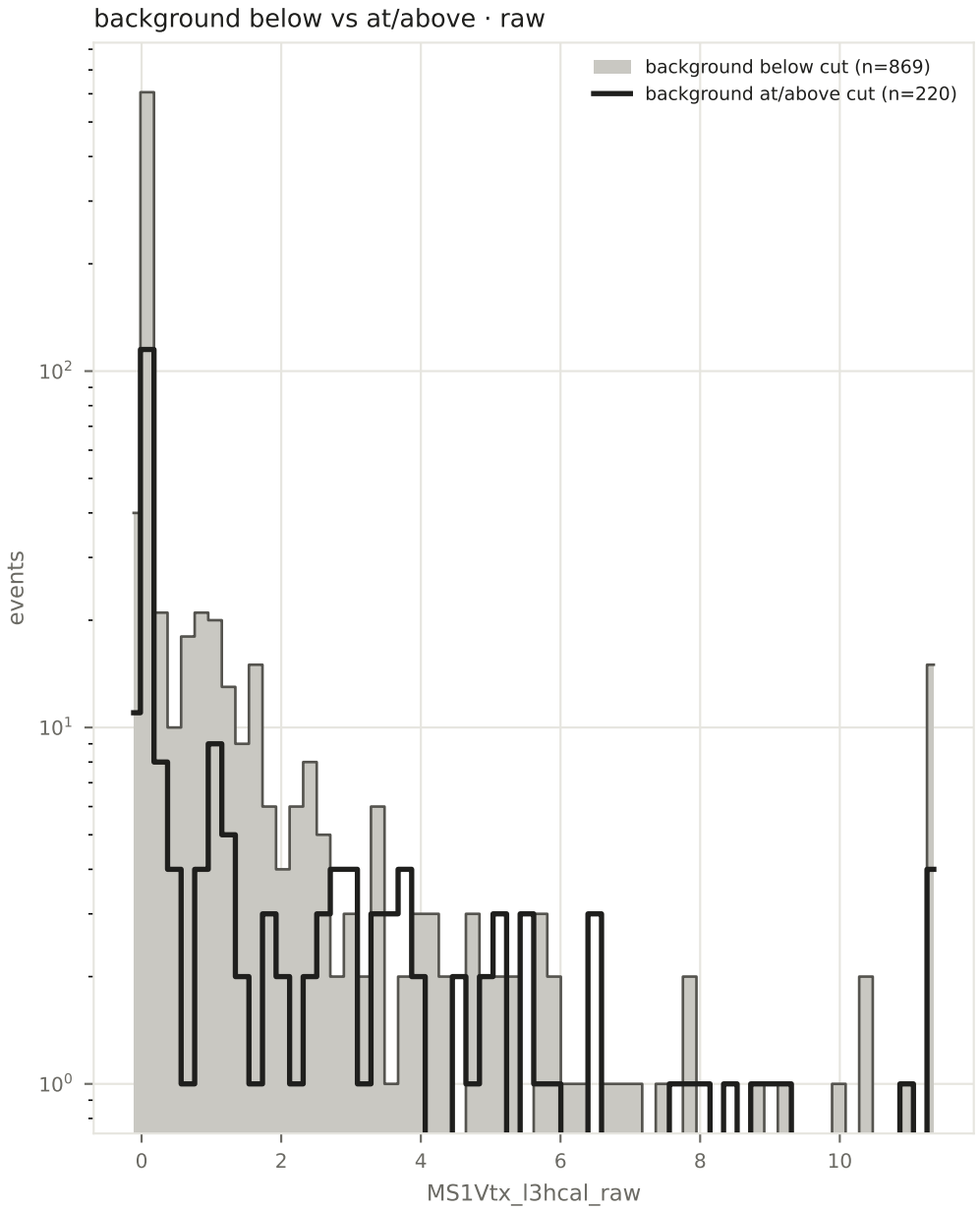


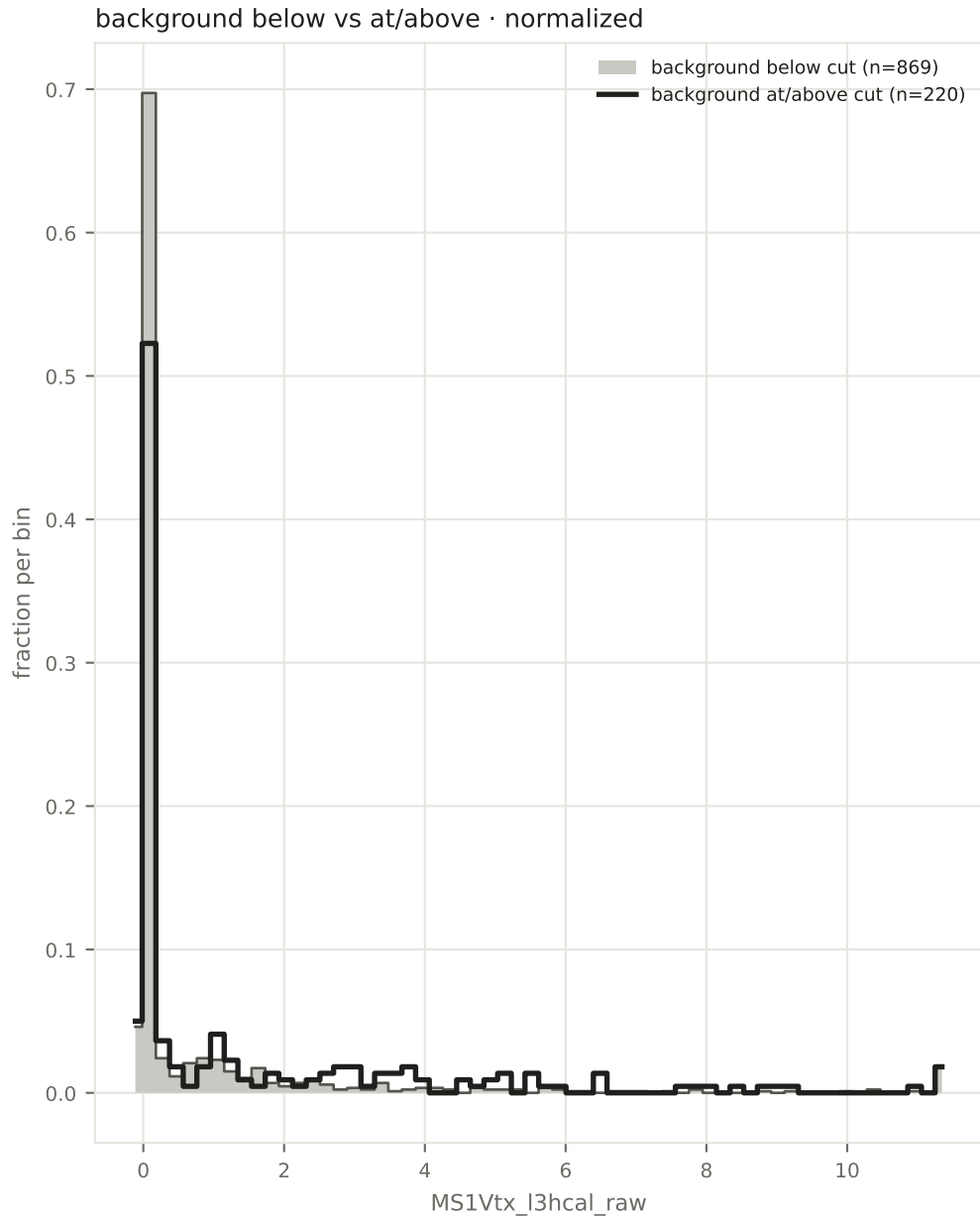

edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_l3hcal_raw

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 95 / 169

D(below vs above) 0.183  n below = 869  n above = 220  large D = background shape changes across the cut



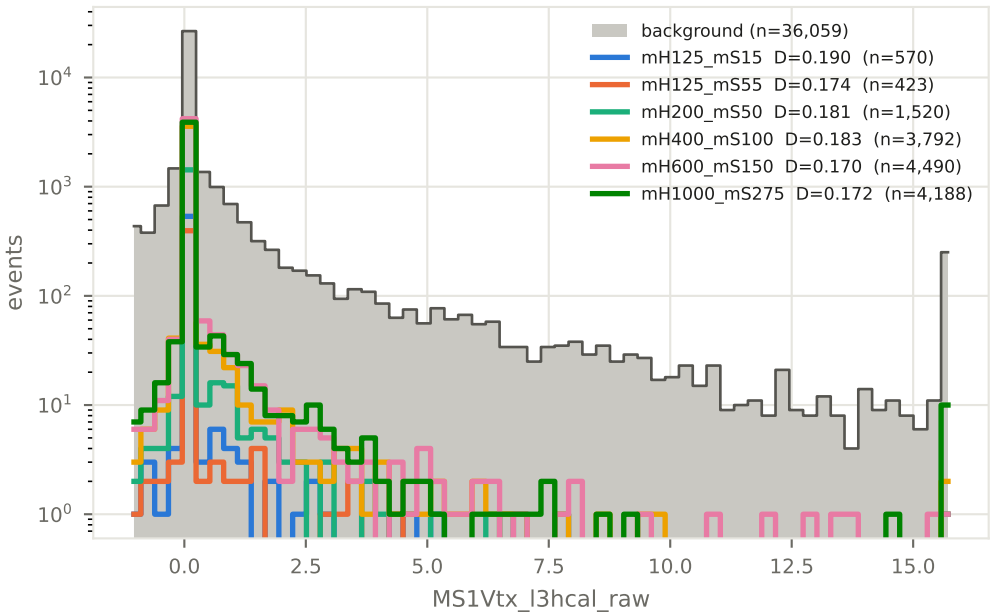



# MS1Vtx_l3hcal_raw

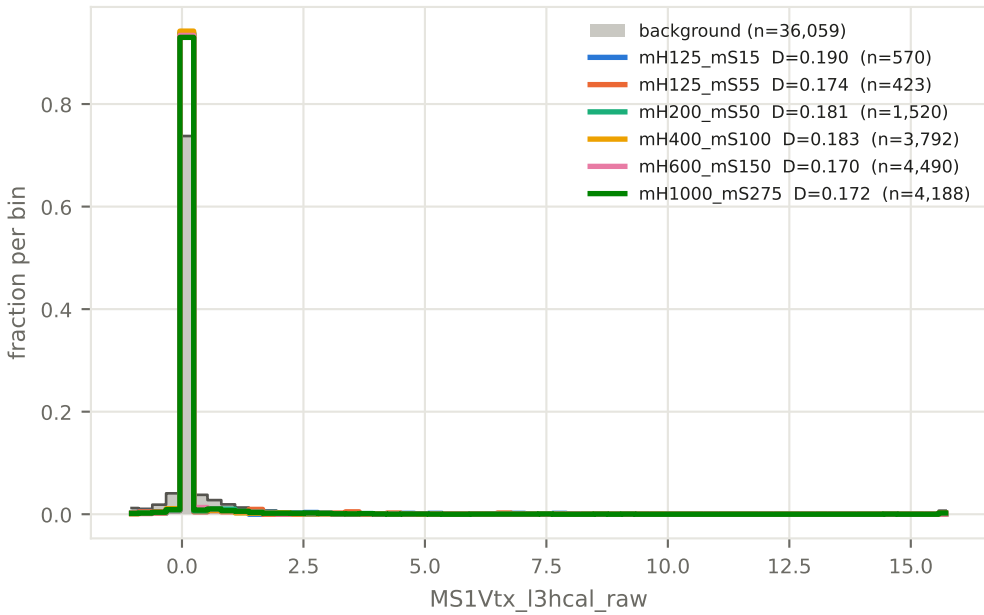
region: ALL · selection: all (every event) · rank 95 / 169

D_equal 0.178 D_pooled 0.176 min 0.170 (mH600_mS150) max 0.190 (mH125_mS15) bkg n = 36,059 sig n = 14,983

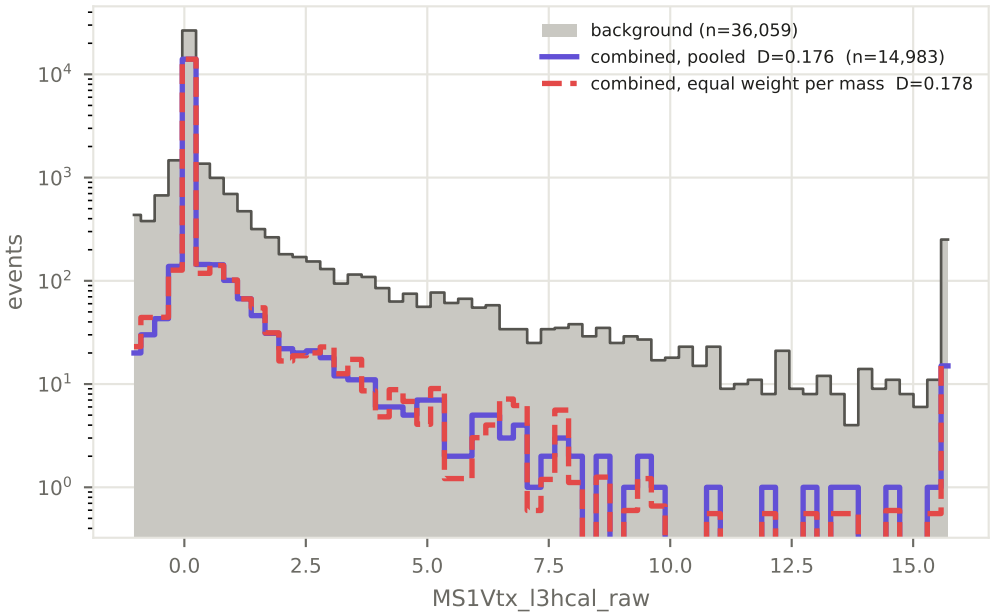
each mass point vs background · raw



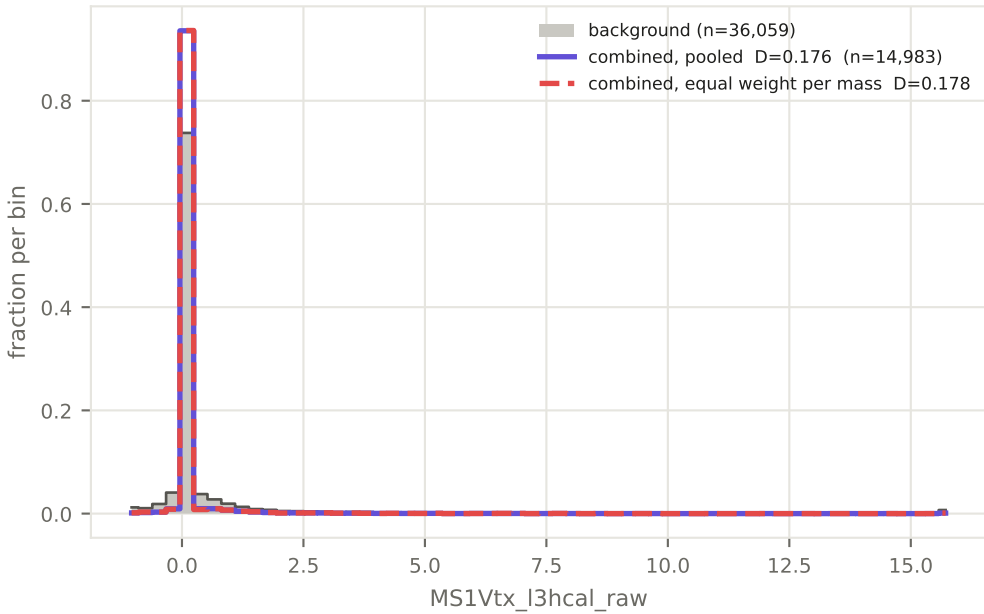

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized



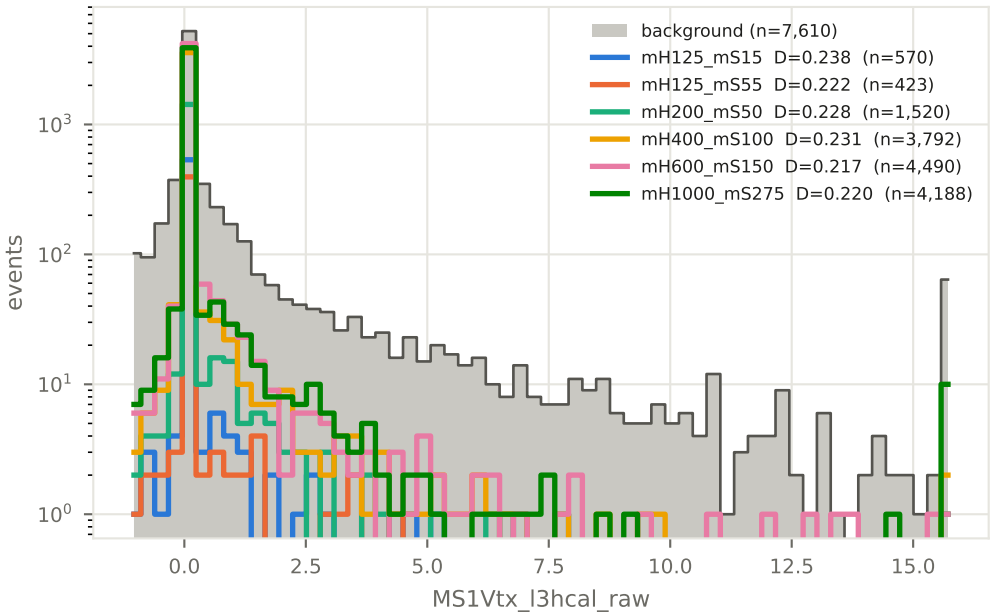

edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_l3hcal_raw

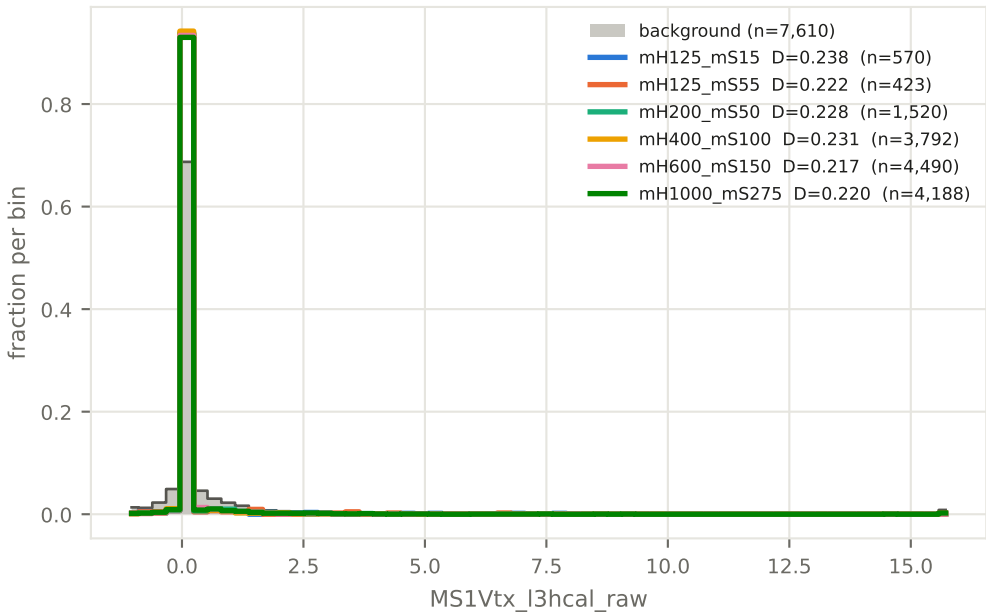
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 95 / 169

D_equal 0.226 D_pooled 0.224 min 0.217 (mH600_mS150) max 0.238 (mH125_mS15) bkg n = 7,610 sig n = 14,983

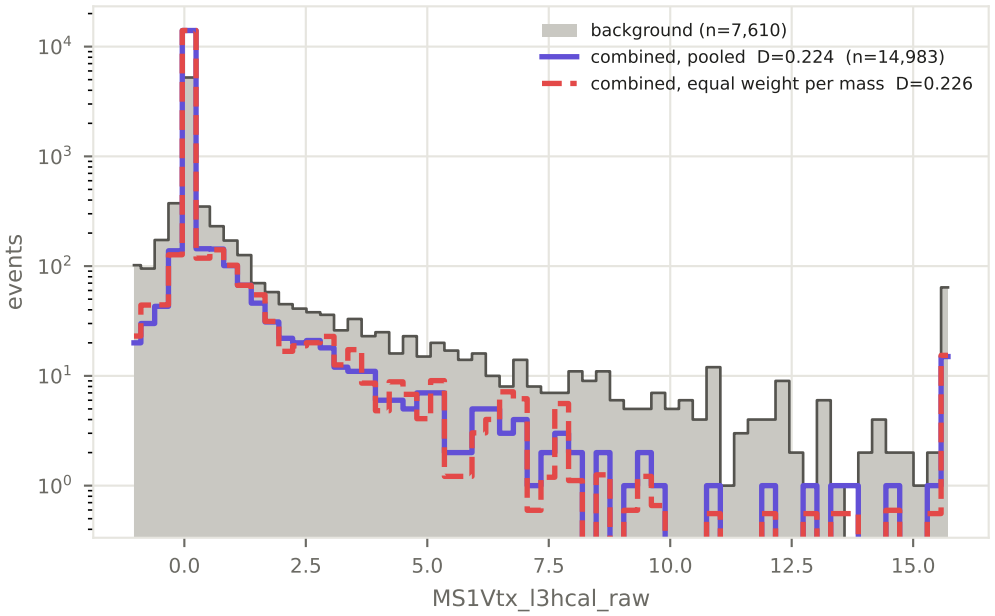
each mass point vs background · raw



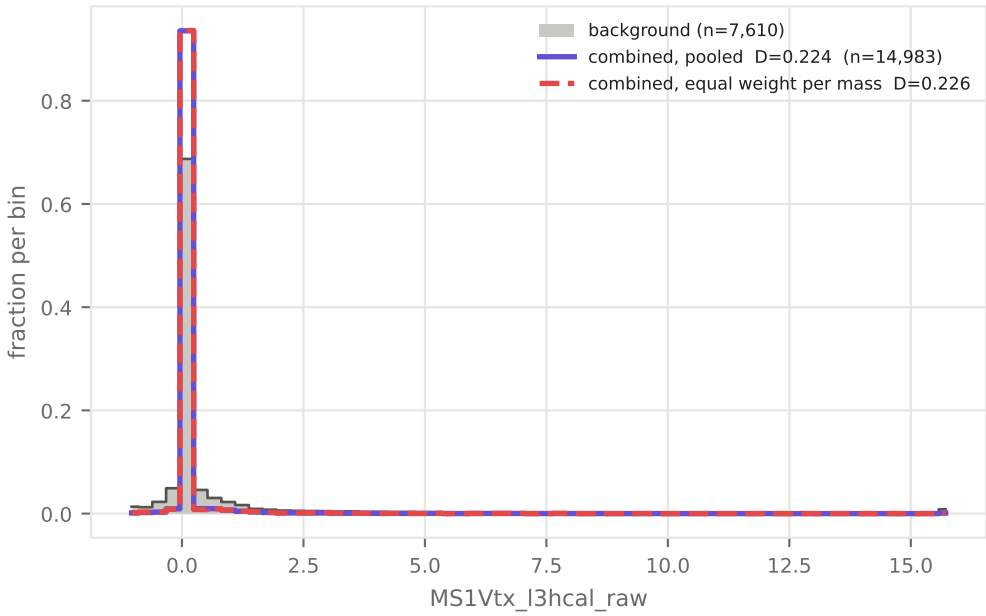

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized




edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_l3hcal_raw

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 95 / 169

D(below vs above) 0.061  n below = 28,449  n above = 7,610  large D = background shape changes across the cut

